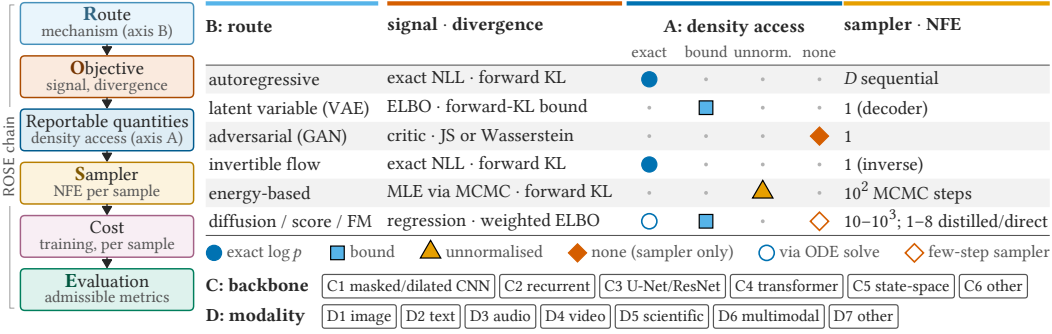
Route
mechanism (axis B)
Objective
signal, divergence
Reportable quantities
density access (axis A)
Sampler
NFE per sample
Cost
training, per sample
Evaluation
admissible metrics
ROSE chain
B: route
signal · divergence
A: density access
exact
bound
unnorm.
none
sampler · NFE
autoregressive
exact NLL · forward KL
D sequential
latent variable (VAE)
ELBO · forward-KL bound
1 (decoder)
adversarial (GAN)
critic · JS or Wasserstein
1
invertible flow
exact NLL · forward KL
1 (inverse)
energy-based
MLE via MCMC · forward KL
10^2 MCMC steps
diffusion / score / FM
regression · weighted ELBO
10–10^3; 1–8 distilled/direct
exact log p
bound
unnormalised
none (sampler only)
via ODE solve
few-step sampler
C: backbone
C1 masked/dilated CNN
C2 recurrent
C3 U-Net/ResNet
C4 transformer
C5 state-space
C6 other
D: modality
D1 image
D2 text
D3 audio
D4 video
D5 scientific
D6 multimodal
D7 other
Composite (latent text-to-image): one (A, B, C, D) tuple per component
solid: trained; dashed: frozen; arrows: inference data flow
prompt
CLIP text encoder
(–, –, C4, D2)
latent denoiser
(bound, diffusion, C3, D1)
VAE decoder
(none, VAE-type, C3, D1)
image